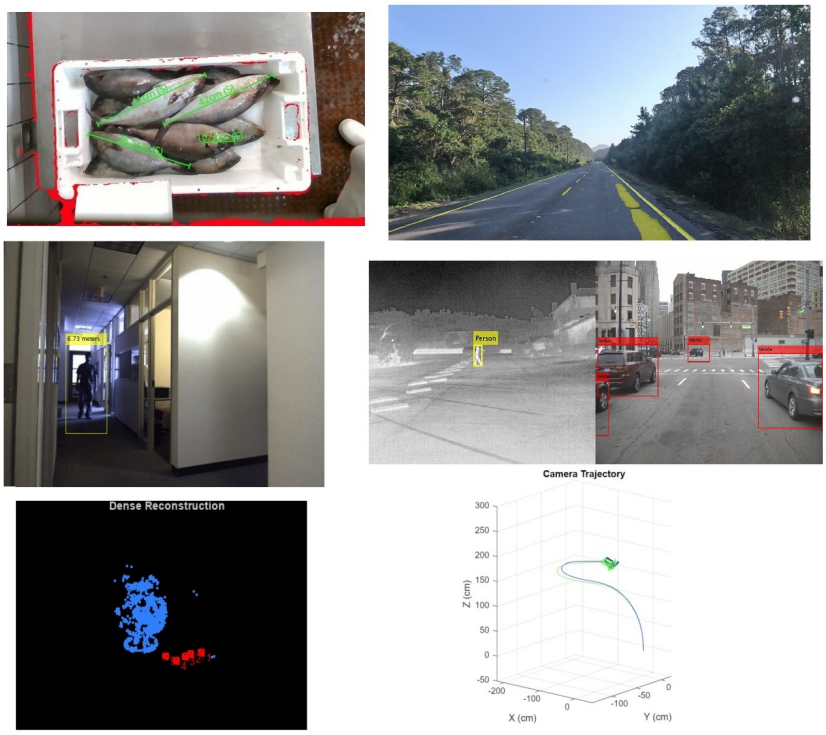
TÉCNICO LISBOA



Computational Vision

Project 2 Guide and Statements

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Project 2 Evaluation Guide

Four different aspects contribute are part of the assignment criteria, each one with the following relative weight in the overall grade:

- Report Structure and Presentation: 4/20
- Report Contents, Implementation and Analysis: 10/20
- Provided Code and Interpretability: 3/20
- Self and peer review: 3/20

In turn, these components are are detailed into two or more parts, accounting for the multiple aspects to be evaluated in the report, as illustrated by Table 1, shown below. The description of each of these sub-components is provided next. It is strongly recommended to carefully review and understand the assessment criteria, not only for for the development of your report, but also to correctly evaluate the ones from other groups.

Table 1: Assessment criteria for Project 2

Report Structure and Presentation		Report Contents, Implementation and Analysis			Provided Code and Interpretability			Self and peer review		Overall
General aspects	Report structure	Key concepts of algorithm	Results, Analysis and Generalisation	Conclusions	General use	User-friendliness code commentaries	Modularity	Self review	Peer review	
2	2	4	5	1	1	1	1	1	2	20

It should be noted that the project final grade will take into account the self and peer review assessments, as well as the evaluation by the faculty.

1. Report Structure and Presentation

Overall weight in grade: 4/20.

- General aspects (2/20):
  - The report should be written in academic English, and obey standard orthographic and grammatical rules.
  - The images generated should be placed at the right of the code in the live script, to avoid unnecessary extension of the document, and to facilitate the follow-up of the code. The images used for contextualization and/or to explain formulas can be placed below the text or at the right.
  - Every image should ne numbered and have a title, using the numbering to mention them throughout report.
  - Together with the live script, the PDF version generated by it is mandatory, to ensure that the evaluation can be made even if an issue with the code occurs.
- Report structure (2/20):
  - The report should be structured using a standard structure (*IMRAD* - Introduction, Methods, Results, Analysis and Developments). More sub-sections or an adequate variation can be used if needed, and conclusions should also be included.

2. Report Contents, Implementation and Analysis

Overall weight in grade: 10/20.

- Key concepts of algorithm (4/20):
  - In the report, the fundamental concepts about the algorithm used for addressing the problem must be explained, depending on the nature of your Project 2. The objective of the project must be clearly and objectively stated. In the case of projects of Stereo Vision, your understanding of the utilized theoretical concepts and how these are used in the provided example must be clear. For the case of projects of Intelligent Vision Systems, the dataset should be characterized and the tools used for the project must be described.
- Results, Analysis and Generalisation (5/20):
  - After describing the central concepts for solving the problem for your project, you should evaluate your solution using the complete dataset (for projects of Intelligent Vision Systems), or for your own images (for projects of Stereo Vision). An analysis should follow regarding the obtained results, and stating the major limitations.

- Conclusions (1/20):
  - Following the analysis regarding the performance of the developed solutions, the conclusions should summarize such results, reflecting on their success, limitations, and pointing to possible directions of improvement as future work.

3. Provided Code and Interpretability

Overall weight in grade: 3/20.

- General use (1/20):
  - A code implementation is suggested since virtually all the tools used for intelligent vision systems and stereo vision in this course are directly available, but Simulink can be used instead, if desired. Nevertheless, if this option is chosen, a live script should still be provided as a report, executing one or more Simulink models which must redirect their output to the live script.
- User-friendliness, code commentaries (1/20):
  - The provided code should be user friendly: the names of the variables should be chosen according to their meaning and use in the code; the code should be commented, explaining the purpose of each function (MATLAB or user-defined) used; and it should be organized by sections, preferably according to the designed algorithms. If a Simulink implementation is used instead, the blocks should be named similarly to their function in the development pipeline, and usage of text boxes for added information is recommended, aiming to have the same purpose as commentaries in a code implementation.
  - Every line of code that is commented is assumed to be a commentary, thus not contributing for the overall code. Therefore, legacy or residual code should be avoided.
- Modularity (1/20):
  - The usage of user-defined functions is strongly recommended, specially when repeating certain methods or tools. For example, if a filtering operation followed by binarization is used several times along the code or for a group of images, these two operations can be defined as a function that takes an input image and a set of parameters (filter intensity, for instance) and returns the filtered and binary image. Functions should be defined in external files and not in the live script, and must be called directly by this script. Naturally, functions should also have code commentaries describing how they work, which inputs they take and the outputs they return.

4. Self and peer review

Overall weight in grade: 3/20.

- Self review (1/20):
  - After submission of the project, you will be asked to grade your own project (limited to 17 out of 20), according to the aforementioned assessment criteria. The capacity to analyse your own work and evaluate it will be checked, and also be a part of the final grade as a self assessment component. Provide an adequate justification of your quantitative assessment with a qualitative analysis regarding positive and to improve aspects of your work.
- Peer review (2/20):
  - The previous task should be repeated for evaluating two reports from your peers, which is done anonymously. Once again, the ability to analyse and grade the projects will be checked, helping you gain an understanding about CV applications other than the one of your project. Please provide the assessed groups a qualitative assessment based on constructive feedback regarding positive and to improve aspects of the work, in agreement with the quantitative assessment you made.
- A detailed Excel sheet for the self and peer review will be provided to each group. The self and peer review grade is based on the quality and completeness of the assessments.

Project 2 Statements

The main objective of this project is to exploit different solutions to achieve the proposed goal. It is not necessary to find a definitive solution, but only to gain insight in how easy/difficult the problem can be. Therefore, the working time dedicated to this work should pay attention to this point and as expected, the final mark of the work will have in consideration more the effort put to exploit approaches to the problem, including feasibility and weak points, than to the quality of the results. It should be noted that this work covers a limited part of the topics of this course, and naturally does not replace the work proposed to the labs. Pay attention that part of the exam questions may be based on the work proposed in the both the projects and the labs.

Report format and software version

The report must be submitted in a ***Live Script***, and it is suggested to become familiar with this tool. Recommended sources for an introduction to *Live Scripts* can be found in the following links: Link 1, Link 2, Link 3. If Simulink is chosen for the implementation, it must be called from the *Live Script* and the resulting image/s routed back to it, so they can be displayed.

All the proposed problems are to be solved using **2023a** and the **correspondent Simulink version**. Other versions of MATLAB may not include some functions, or these may vary in the input/output arguments, and therefore all the delivered files, namely the live script, Simulink models and/or additional functions should run using this version.

Submission instructions

The project must be submitted in Fenix as a zip file, having inside the MATLAB 2023a live script file and any other MATLAB 2023a or required files to run it. If, for any reason, you wish to join a film or a large set of images, those files may be emailed via link with unlimited download time. The name of the project zip file to be submitted in Fenix should follow the pattern:

G<group#>_<student#1>_<student#2>_<student#3>.zip
e.g., “G10_11111_22222_33333.zip”.

Submission deadline: 3/4/2026

Self and peer review guidelines

Once the submission deadline has passed, you will receive an Excel table with fields corresponding to the multiple assessment criteria. You will then evaluate your own report accordingly, as well as the ones from two other groups. As stated before, the ability to analyse and evaluate the reports from other groups, as well as your own, is also part of your own evaluation.

Self and peer review deadline: 10/4/2026

Intelligent Vision Systems

Topic A1: Automated image analysis in fish counting for marine research

Deep learning has many applications that aim to simplify tasks in various domains, from healthcare to environmental monitoring. By automating complex processes, AI-driven vision-based solutions enhance efficiency, reduce human effort, and improve accuracy in data-driven tasks. One such application is in monitoring fish populations resorting to data recovered from fisheries, where manual fish counting can be time-consuming and prone to errors.

For Project 2, this topic focuses on the automated quantification of fish in storage boxes as shown in Figure 1 using the image-based methods addressed in the Intelligent Vision Systems part of the course. By leveraging deep learning techniques, the goal is to accurately count fish from the provided dataset. Beyond simple counting, the objective could extend to fish weight and size estimation, contributing to scientific research, population assessment, and sustainable resource management.

The usage of Machine/Deep Learning tools is required, and helpful examples can be found in these two links: image categorization example, image segmentation example. Additional examples of the Statistics and Machine Learning Toolbox and/or Deep Learning Toolbox may be of use, so feel free to explore them, but do not forget to adequately cite the source.

Figure 1: Sample image from the dataset of Topic A1.

Problems A1.1 and A1.2

For the given dataset (available here), it is intended to:

1. Analyse the images, characterize the available dataset and split it into folders accounting for the number of fish present in each. Each folder, and respective name, will correspond to a class: for example, 5 classes of ”1”, ”2”, ”3”, ”4” or ”more” fishes. Be careful not to rename them.

2. Employ tools of Intelligent Vision Systems to automatically count the fish in the provided images of the dataset. It is suggested that you start with a simple architecture for the neural network.

3. Evaluate your classifier for the available dataset, and analyse the obtained results with a systematic procedure.

4. Try to improve your algorithm by exploring the additional options MATLAB provides for the structure of the neural network and training options.

Intelligent Vision Systems

Topic A2: Semantic segmentation for perception in autonomous driving

Deep learning has become a key technology in enabling intelligent perception systems across a wide range of applications, including robotics, transportation, and smart infrastructure. By leveraging large annotated datasets and powerful neural network architectures, modern computer vision systems can interpret complex visual scenes and extract meaningful information from images. One particularly important application is in autonomous driving, where vehicles must continuously analyze their surroundings to identify roads, obstacles, and other relevant elements of the environment.

For Project 2, this topic focuses on the semantic segmentation of road scenes using the FCCD Dataset (respective scientific article available here), which contains images from autonomous driving scenarios with pixel-level annotations, as shown in Figure 2. The objective is to develop a model capable of segmenting the semantic classes present in the dataset images, contributing to perception systems used in autonomous driving.

The usage of Machine/Deep Learning tools is required, and helpful examples can be found in these two links: image categorization example, image segmentation example. Additional examples of the Statistics and Machine Learning Toolbox and/or Deep Learning Toolbox may be of use, so feel free to explore them, but do not forget to adequately cite the source.

Figure 2: Sample image from the dataset of Topic A2.

Problems A2.1 and A2.2

For the given dataset (available here), it is intended to:

1. Analyse the images and annotations, characterize the dataset, and identify the semantic classes present. Organize the data for training, validation, and testing.

2. Implement a semantic segmentation pipeline using tools from Intelligent Vision Systems, starting with a simple neural network architecture.

3. Train the model and evaluate its performance using appropriate metrics, analysing the obtained results systematically.

4. Improve the model by exploring different architectures, training parameters, data augmentation techniques, or preprocessing strategies available in MATLAB.

Intelligent Vision Systems

Topic A3: People and vehicle detection using RGB and thermal imagery

Deep learning has significantly improved the capabilities of computer vision systems in real-world perception tasks. In safety-critical applications such as autonomous driving and advanced driver-assistance systems (ADAS), reliable detection of pedestrians and vehicles is essential for safe navigation. Vision-based systems enable automated analysis of road environments, allowing vehicles to identify relevant objects and react accordingly.

For Project 2, this topic focuses on detecting people and vehicles using the FLIR ADAS dataset, which contains RGB and thermal images captured in real driving scenarios as shown in Figure 3, representing a fundamental task for autonomous driving and advanced driver-assistance systems (ADAS). The objective is to develop a model capable of automatically detecting pedestrians and vehicles in the dataset images.

The usage of Machine/Deep Learning tools is required, and helpful examples can be found in these two links: image categorization example, image segmentation example. Additional examples of the Statistics and Machine Learning Toolbox and/or Deep Learning Toolbox may be of use, so feel free to explore them, but do not forget to adequately cite the source.

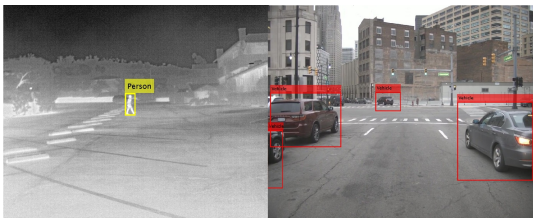

Figure 3: Sample image from the dataset of Topic A3.

Problems A3.1, A3.2, A3.3 and A3.4

For the given dataset (available here), it is intended to:

1. Choose to work using either the RGB (A3.1 and A3.2) or the thermal imagery (A3.3 and A3.4), characterize the selected dataset, and identify the relevant object classes (e.g., pedestrians, cars, bicycles, or other vehicles). Prepare the dataset for training, validation, and testing.

2. Implement an object detection pipeline using tools from Intelligent Vision Systems, starting with a simple detection architecture.

3. Train and evaluate the model using appropriate metrics, and analyse the obtained results.

4. Improve the detection system by exploring different architectures, training parameters, preprocessing techniques, or data augmentation strategies. Optionally compare results using RGB and thermal images.

Stereo Vision

Topic B1: Monocular visual odometry

Stereo and multiview vision techniques enable the estimation of camera motion and scene structure from image sequences. These methods are widely used in applications such as mobile robotics, autonomous driving, and unmanned aerial vehicles, where visual information can be used to estimate the movement of a camera over time.

For Project 2, this topic focuses on monocular visual odometry using the tools provided in the MATLAB Computer Vision Toolbox. Visual odometry estimates the trajectory of a moving camera by analysing a sequence of images and tracking visual features across frames. The objective is to understand the principles behind visual odometry and estimate the trajectory of a camera from a sequence of images.

Students are encouraged to explore the MATLAB example available in this link. Additional examples and documentation from the Computer Vision Toolbox may also be useful, but do not forget to adequately cite the source.

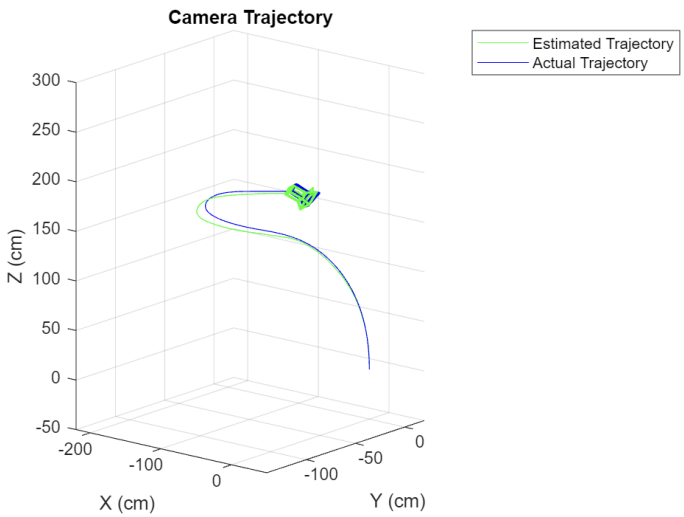

Figure 4: Camera trajectory reconstructed using visual odometry.

Problems B1.1 and B1.2

For this problem, it is intended to:

1. Follow the MATLAB example and understand the main mathematical principles of the applied methods.

2. Study the movement of translation in a fixed direction and determine the scale factor using a calibration object.

3. Acquire your own image sequences using a camera (e.g., a laptop or webcam) and analyse the obtained trajectory. If possible, validate the results using the Robotics Arena apparatus or another experimental setup.

Stereo Vision

Topic B2: Depth estimation from stereo video

Stereo vision enables the estimation of depth by analysing two images captured from slightly different viewpoints. By identifying corresponding points in the left and right images, it is possible to compute disparity and estimate the distance to objects in the scene. These techniques are widely used in robotics, autonomous driving, and 3-D perception systems.

For Project 2, this topic focuses on estimating depth from stereo video using the MATLAB Computer Vision Toolbox. The objective is to estimate distances to objects in a scene using stereo image pairs and analyse the obtained depth measurements.

Students are encouraged to explore the MATLAB example available in this link. Additional examples and documentation from the toolbox may also be useful, but do not forget to adequately cite the source.

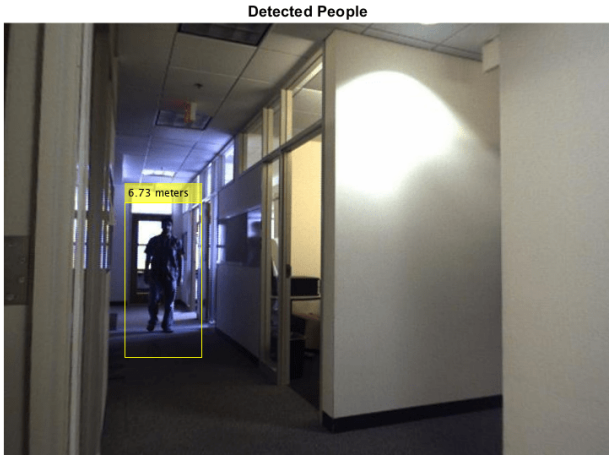

Figure 5: Distance measurement with stereo video.

Problems B2.1 and B2.2

For this problem, it is intended to:

1. Follow the MATLAB example and understand the mathematical principles of stereo depth estimation.

2. Apply the method to your own stereo images or videos. You may use people as an example, but at least one additional object should be analysed.

3. Evaluate the estimated distances and, if possible, validate some results using the Robotics Arena apparatus or another experimental setup.

Stereo Vision

Topic B3: Structure from motion from multiple views

Structure from Motion (SfM) is a computer vision technique used to estimate the three-dimensional structure of a scene from multiple two-dimensional images captured from different viewpoints. These methods are widely used in applications such as robot navigation, augmented reality, and 3-D reconstruction.

For Project 2, this topic focuses on reconstructing the 3-D structure of a scene using multiple images and the tools available in the MATLAB Computer Vision Toolbox. The objective is to estimate camera poses and reconstruct the 3-D structure of a scene from a sequence of images.

Students are encouraged to explore the MATLAB example available in this link. Additional examples and documentation from the toolbox may also be useful, but do not forget to adequately cite the source.

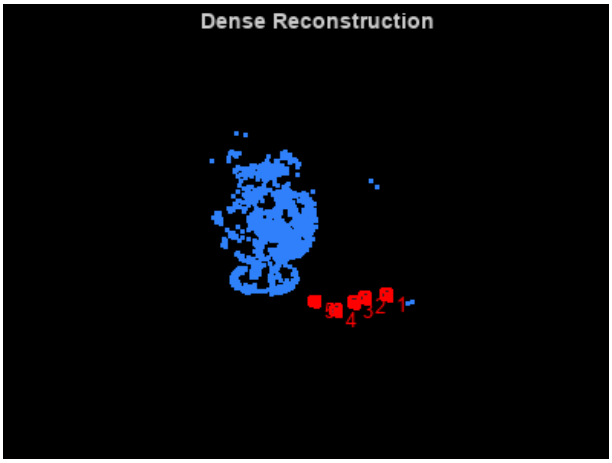

Figure 6: Reconstruction of an object using structure from motion.

Problems B3.1 and B3.2

For this problem, it is intended to:

1. Follow the MATLAB example and understand the mathematical principles of the structure-from-motion pipeline.
2. Apply the method to reconstruct your own objects with known dimensions and discuss the obtained results.
3. Experiment with different cameras (e.g., a phone camera) and analyse the impact on the reconstruction results.